Amitesh Singh

Backend & Real-time Systems Engineer · Exploring AI Agents

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SUMMARY

Backend-focused engineer who builds production systems end to end: an exchange with an in-memory matching engine, real-time multiplayer games on raw WebSockets, and autonomous AI agents. Former founding engineer at an EV mobility startup, where I shipped the product, set up CI/CD, and hired the engineering team.

EXPERIENCE

Macro Rides · zone-based EV ride platform, Delhi — Oct 2024 – Mar 2026
Founding Full Stack Engineer

- First engineer; architected and shipped the platform from zero to production across frontend, backend, and DevOps.
- Replaced a ~30-minute manual deploy with a one-push CI/CD pipeline.
- Ran technical hiring end to end: designed interviews, evaluated candidates, and onboarded 2–3 engineers.
- Owned production reliability and engineering timelines through the early-rider launch.

My Design Minds · design & manufacturing platform for MSMEs — Apr 2025 – Nov 2025
Full Stack Engineer (Contract, part-time)

- Built the AI-driven product from the ground up, leading frontend, backend, and DevOps for the MVP.
- Built the production AI inference flows that power the core product.
- Managed and mentored a 2-person engineering team.

PROJECTS

Centralized Exchange | *TypeScript, Bun, Redis, PostgreSQL, WebSockets, Docker* — Live | GitHub

- Built an exchange around an in-memory order-matching engine, decoupled from Postgres via Redis so matching never blocks on I/O.
- Split into independently scalable services (HTTP API, engine, WebSocket server, DB worker) linked by Redis pub/sub.
- Streamed live order books, prices, and candlestick charts to clients; deployed with Docker, Nginx, and GitHub Actions.

QuickDuel | *WebSockets, Redis, BullMQ, PostgreSQL, Next.js, AWS* — Live | GitHub

- Real-time multiplayer maths game with server-authoritative state over WebSockets keeping both players in sync.
- Queue-based matchmaking in Redis to avoid race conditions; BullMQ workers for match and result jobs.

Browser Agent | *TypeScript, Playwright, OpenAI API, Express, Prisma, Next.js* — GitHub

- Autonomous agent that completes natural-language tasks on live websites with an observe → plan → act loop driving Chromium. Built at AI Hackathon S30 3.0.

AI Agent CLI | *TypeScript, Bun, OpenAI & Gemini APIs, E2B* — GitHub

- Terminal coding agent (a mini Claude Code) with a ReAct loop, tool calling, human-in-the-loop approvals, and a sandbox.
- Persistent cross-session memory, automatic context summarization, and pluggable model providers.

tui-typer | *TypeScript, React, Ink, WebSockets* — npm | GitHub

- Multiplayer terminal typing race with real-time rooms; published on npm with 1,000+ downloads in the last year.

SKILLS

Languages: TypeScript, JavaScript, SQL
Backend & Real-time: Node.js, Bun, Express, WebSockets, Redis (queues, pub/sub), BullMQ, PostgreSQL, Prisma
AI: LLM agents, tool calling, OpenAI & Gemini APIs, Playwright browser automation, RAG, E2B sandboxes
DevOps: Docker, AWS (EC2), Nginx, GitHub Actions CI/CD, Turborepo
Frontend: React, Next.js, Tailwind CSS, Canvas API, Ink

EDUCATION

100xSchool · Super 30 (engineering cohort) — May 2026 – Present
University of Delhi · B.Com, School of Open Learning — 2023 – 2026